Suppression of the quadrupole-misalignment false EDM signal by orbit-based methods in a proton EDM storage ring

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In the storage ring proposed for measuring the proton electric dipole moment (EDM), quadrupole misalignments of a few microns mimic a genuine EDM through a geometric (Berry) phase, at up to a thousand times the target sensitivity of $10^{-29}\, e$cm. The reference design controls this with beam-based alignment (BBA) together with a four-fold measurement — the counter-rotating–beam difference taken at both quadrupole polarities — that cancels the effect when the machine is identical in the two polarity states. Using symplectic spin tracking we ask how far standard orbit correction alone can go. The false EDM depends on two distinct parts of the misalignment: an antisymmetric part that distorts the closed orbit, and a symmetric part to which the orbit barely responds; the second is the obstacle, and the orbit inversions and amplitude readouts that would reconstruct it from a single beam all fail (null-finding BBA reaches it, but slowly and invasively). We show that the two counter-rotating beams, whose orbit responses are nearly opposite, carry the symmetric part in the *sum* of their orbits — invisible in either beam and in their difference, the separation the reference chain monitors, but fully exposed in the sum. Correcting both orbits jointly therefore reaches the symmetric component from the orbit alone: against a calibrated reference, on a raw 10 μm machine at a realistic 1% β-beat, the counter-rotating difference falls to a median 0.043 times the target over fifteen seeds, all sub-target, and is unchanged by β-beating and unmodeled quadrupole roll. The same sum, however, carries the static BPM offset, so symmetric misalignment and offset are degenerate there: absolute alignment still needs one calibration campaign (BBA or the four-fold), and this we make precise. What two-beam correction removes is the *repeated* campaign: the drift is differential, the static offset cancels in the orbit change, and a continuous two-beam feedback holds the symmetric drift — invisible to single-beam feedback — at the offset-free level. Its role is thus a continuous symmetric-drift monitor that maintains an initial alignment indefinitely, in place of the periodic re-alignment the reference program requires. We verify the four-fold cancellation as the baseline and give the resulting operating requirements.

I. INTRODUCTION

A. The frozen-spin method and the false-EDM problem

A permanent electric dipole moment (EDM) of the proton would violate time-reversal symmetry, and its observation above the Standard-Model expectation ($\lesssim 10^{-31}\, e$cm) would constitute direct evidence of new sources of CP violation. The proposed storage-ring experiment [1] targets a sensitivity of $d = 10^{-29}\, e$cm using the *frozen-spin* method: protons are stored at the "magic" momentum ($p \approx 0.7$ GeV/c), at which the spin remains aligned with the momentum as the beam circulates. An EDM then reveals itself as a slow rotation of the spin out of the ring plane, driven by the radial electric bending field. The measured quantity is the out-of-plane precession rate,

$$f \equiv \frac{dS_y}{dt}, \qquad d = 10^{-29}\, e\cdot\text{cm} \;\leftrightarrow\; f \approx 1 \text{ nrad/s}. \tag{1}$$

Any competing effect that rotates the spin out of plane at this level must be eliminated or bounded; such effects are referred to as *false EDM* signals.

The subject of this paper is the false EDM generated by transverse misalignments of the focusing quadrupoles, a systematic studied for an all-electric version of the ring in Ref. [2] and central to the error budget of Ref. [1]. A quadrupole has zero field on its magnetic axis, so a beam passing off-axis experiences a field proportional to the offset. A vertical offset dy exposes the beam to a radial magnetic field, which precesses the spin about the radial axis and can therefore tilt it out of the ring plane. A horizontal offset dx exposes the beam to a vertical field, which precesses the spin about the vertical axis; this precession is confined to the ring plane and, like the in-plane precession of the frozen-spin condition itself, does not accumulate in S_y. Either precession alone is nearly harmless: averaged around the ring, its out-of-plane effect largely cancels. When both are present, however, the two precessions do not commute — precessing first about the radial axis and then about the vertical axis is not the same operation as the reverse — and the residue of this non-commutativity accumulates turn after turn as a net rotation about the third axis, which is precisely the out-of-plane direction in which the EDM is sought. This is a geometric (Berry) phase, and its leading dependence is

$$f_{\text{false}} \;\propto\; dx \cdot dy \;\Longrightarrow\; f_{\text{false}} \propto \sigma^2 \tag{2}$$

for misalignments of rms size σ. We verify the quadratic scaling directly in Sec. II B (measured exponent $p = 2.00 \pm 0.01$, in agreement with Fig. 9(a) of Ref. [1]). At a realistic alignment of $\sigma = 10\ \mu$m the false EDM is of order a thousand times the target of Eq. (1), and it appears in the same observable, dS_y/dt, as the genuine signal.

B. The ring

The lattice studied here is the symmetric-hybrid ring of Ref. [1]: electric bending with magnetic focusing. The ring ($R_0 = 95.49$ m) consists of 24 identical FODO cells. Each cell contains a horizontally focusing quadrupole (QF) and a horizontally defocusing quadrupole (QD) — both *magnetic*, with gradients of equal magnitude ($g \approx 0.2$ T/m) and opposite sign — separated by field-free drifts and by the *electric* bending deflectors:

$$\mathrm{QF} \to \mathrm{drift} \to \mathrm{deflector} \to \mathrm{drift} \to \mathrm{QD}$$
$$\to \mathrm{drift} \to \mathrm{deflector} \to \mathrm{drift} \to (\mathrm{QF}\ldots)$$

The deflectors are cylindrical electrostatic capacitors that bend the beam in the horizontal plane; at the magic momentum ($p = 700.7$ MeV/c, $\gamma = 1.248$, $\beta = 0.598$) their radial field also freezes the in-plane spin precession. The radial field set by the magic condition and the ring radius is $E_0 = \beta c\,(p/qR_0) \approx 4.4$ MV/m (the value used in the tracker; the local field within the deflectors, which occupy a fraction of the circumference, is correspondingly larger). Two properties of this arrangement matter for what follows. First, between any two neighboring quadrupoles the beam traverses a deflector: the deflectors bend and weakly focus the beam *horizontally*, while in the *vertical* plane they act, to a very good approximation, as field-free drifts. Second, the betatron tunes, measured from the intra-turn betatron oscillation in the tracker with the deflectors included, are $Q_x = 2.64$ and $Q_y = 2.30$ (the tune is the number of transverse oscillations per revolution) — close to the design values 2.70 and 2.245 of Ref. [1], which this lattice is modeled on but does not reproduce exactly. The planes are *not* degenerate: the horizontal focusing of the deflectors raises Q_x well above Q_y, whereas an analytic model that treats the deflectors as drifts in both planes gives 2.30 for each and misses the split (a fact that returns in the BBA steering of Sec. IV B). It is the vertical tune $Q_y \approx 2.3$ that sets the orbit gain law of Sec. III A, since the false EDM is driven through the vertical plane.

C. The proposed control program and the question of this paper

The design study of Ref. [1] already specifies a program to control this systematic, with two ingredients. The first is *beam-based alignment* (BBA): the accelerator community locates individual quadrupole centers by varying the quadrupole gradient and finding the beam position at which the orbit stops responding, a procedure accelerated with ac excitation [5] and parallelized over many magnets [6]. The second is a *four-fold measurement*: the genuine EDM is odd under beam reversal, so the counter-rotating difference of the two stored beams isolates it from common-mode systematics [Eq. (C1) of Ref. [1]], and repeating that difference with the polarity of all focusing quadrupoles reversed [its Eq. (C2)] cancels the magnetic-lattice false EDM exactly — provided the machine presents the same misalignment in both polarity states. This four-fold cancellation is powerful, but it is not free: the two polarity states cannot be stored at once, so the experiment must be run a second time at reversed polarity, and, as we show in Sec. IV C, the cancellation is undone by any change of alignment between the two runs.

Against this backdrop the natural question is what *standard orbit correction* — flattening the closed orbit with steering elements, guided by the singular-value decomposition of the response matrix, routine since the work of Chung, Decker, and Evans [3], and continuous and non-invasive during data taking — can contribute. The closed orbit is read by beam position monitors (BPMs) at each of the 48 quadrupoles; related tools include orbit-response modeling under degeneracy [4], symmetry correction [7], and on-line response-matrix refinement [8]. None of these techniques is new in itself. What we examine is how each performs on the specific misalignment component that drives the false EDM in this ring, and the result is a reversal of the natural expectation: while no single-beam orbit measurement reaches that component, correcting the two counter-rotating beams jointly does, because the symmetric misalignment lives in the sum of the two orbits. That same sum carries the static BPM offset degenerate with it, so the two-beam correction does not displace the one-time absolute alignment — but it does displace the *repeated* one, maintaining the symmetric alignment against drift, which a single-beam feedback cannot.

D. Findings and outline

Three results organize the paper.

First, *the false EDM has a structure that a single alignment tolerance cannot capture* (Sec. III). It is a bilinear functional of the horizontal and vertical misalignment patterns, and each pattern separates into an *antisymmetric* component (neighboring QF and QD displaced oppositely), which produces a large orbit distortion, and a *symmetric* component (QF and QD displaced together), whose orbit signature is suppressed by up to two orders of magnitude mode by mode. At fixed rms σ the false EDM varies by more than an order of magnitude with which component the misalignment carries.

Second, *the reference solution reaches the target but at an operational cost* (Sec. IV). Beam-based alignment reaches the symmetric component and, iterated far enough, the target on its own (eight passes in our tracking), though slowly; the four-fold polarity measurement of Ref. [1] cancels the false EDM in one step instead. We reproduce both. The cost is operational: the four-fold requires two sequential polarity states in which the machine presents the same symmetric alignment, and a micron-scale symmetric drift between them, invisible to the orbit, re-opens the false EDM.

Third, *counter-rotating orbit correction reaches the symmetric component from the orbit, and its natural role is a continuous symmetric-drift monitor* (Sec. V). A single-beam orbit measurement cannot reach the symmetric component: it is nearly invisible to one beam, and the inversion that would recover it is ill-conditioned. But the two counter-rotating beams have nearly opposite orbit responses, so the symmetric component — invisible in either beam and in their difference, the separation the reference chain monitors — is fully exposed in their *sum*. Correcting both orbits jointly against a calibrated reference therefore reaches it, below target at realistic optics error and roll. The same sum, however, carries the static BPM offset, degenerate there with the symmetric misalignment, so absolute alignment still needs one calibration campaign. What the two-beam correction removes is the *repeated* one: the drift is differential, the offset cancels in the orbit change, and a continuous two-beam feedback maintains the symmetric alignment against drift — the component a single-beam feedback cannot hold — indefinitely.

Section VI examines the operational side — alignment drift, the re-alignment cadence, and the time budget — and Sec. VII collects the conclusions. The simulation infrastructure and the spin-signal estimator, on which all quantitative statements rest, are described first in Sec. II.

II. SIMULATION METHOD

A. Tracking tools

All results are obtained with a particle-and-spin tracking code that integrates the equations of motion element by element through the fields of the lattice of Sec. I B, using a fourth-order Gauss–Legendre symplectic integrator; the spin is propagated in the same steps by the Thomas-BMT equation. Symplectic integration prevents the slow accumulation of energy and amplitude errors, which is a prerequisite for resolving precession rates at the nrad/s level over millions of turns. The genuine EDM enters the model through a switch; with the coupling set to $\eta = 1.88 \times 10^{-15}$ the code reproduces $dS_y/dt = 9.81 \times 10^{-10}$ rad/s, odd under beam reversal, in agreement with Eq. (C1) of Ref. [1]. All false-EDM results below are obtained with the EDM switch off, so that any measured out-of-plane precession is by construction a systematic effect. Quadrupole misalignments, gradient errors (β-beating), and roll angles enter the tracker directly as element parameters.

Wide parameter scans (noise Monte Carlo, β-beating sweeps) use a second, independent layer: the lattice functions, tunes, and the orbit response matrix,

$$R_{ij} = \frac{\sqrt{\beta_i\beta_j}}{2\sin\pi Q}\,(KL)_j\cos\bigl(|\phi_i-\phi_j|-\pi Q\bigr), \qquad (3)$$

evaluated in closed form from the quadrupole and drift transfer matrices. This analytic layer runs in seconds, agrees with the tracker at the percent level in the vertical plane, and shares nothing with it except the configuration file, so that agreement between the two cannot arise from a common implementation error. Two of its limitations play a role later and are recorded here: it treats the electric deflectors as field-free drifts, which is accurate vertically but not horizontally (Sec. IV); and an idealized response matrix can misrepresent the real beam dynamics, so wherever a response matrix enters quantitatively — both the inverse problems of Sec. V B and the steering of the BBA of Sec. IV — it is rebuilt from tracking, and the fidelity this demands is itself one of our results (Sec. IV B).

B. Measuring the false EDM

The quantity of interest is the secular slope of $S_y(t)$. Extracting it reliably requires care, because the betatron oscillation of the tracked particle feeds oscillatory components into the spin motion that exceed the sought slope by orders of magnitude. Two measures are combined.

First, the tracked particle is placed on the closed orbit, where no betatron oscillation is present. The misalignment-distorted closed orbit is found by Newton iteration in a few steps.

Second, the spin history is fit with the model $S_y(t) = a + bt + \sum_k [c_k\cos\omega_k t + d_k\sin\omega_k t]$, and only the secular slope b is retained; residual oscillatory content is absorbed by the harmonic terms rather than biasing the slope, as it would in a plain polynomial fit.

The estimator was validated in three independent ways: (i) scanning $\sigma = 2.5$–10 μm yields $|f| \propto \sigma^p$ with $p = 2.00 \pm 0.01$, the pure quadratic scaling of a geometric phase with no linear leakage; specifically, at $\sigma = 5$ μm after orbit correction we measure $f = 0.62 \times 10^{-9}$ rad/s (median of 3 random seeds) compared to 2.72×10^{-9} rad/s at $\sigma = 10$ μm, giving an observed ratio of 4.36 in agreement with the σ^2 prediction of 4.0; (ii) under sign flips of the misalignment pattern the measured f obeys $f(+dx,-dy) = -f(+dx,+dy)$ and $f(-dx,-dy) = +f(+dx,+dy)$ at the 10^{-3} level, confirming the bilinear character of Eq. (2); (iii) the average of four particles launched symmetrically around the closed orbit reproduces the single on-orbit result within 1%.[1]

[1] The symmetric four-particle average must be constructed around the closed orbit; constructed around the ideal axis it retains the common betatron component induced by the orbit offset itself and fails by orders of magnitude.

III. THE FALSE EDM: MECHANISM AND CHANNELS

The false EDM of Eq. (2) is set by which part of the misalignment the pattern carries, not by its rms alone. This section makes that precise: it separates each misalignment plane into a symmetric and an antisymmetric component with sharply different orbit visibility, and shows that the false EDM is a bilinear form in the two planes whose four channels are dominated by the antisymmetric factors. This structure — established here without reference to any correction scheme — fixes what every later method must contend with: the antisymmetric part is easy to see and to remove, the symmetric part is neither.

A. Symmetric and antisymmetric misalignments

Let the 48 offsets of one plane form a vector v, with QF at index $2k$ and QD at index $2k{+}1$ in cell k. Within each cell we relabel the two offsets by their common part and their difference,

$$m_k = \tfrac{1}{2}\big(v[2k] + v[2k{+}1]\big), \qquad d_k = \tfrac{1}{2}\big(v[2k] - v[2k{+}1]\big), \tag{4}$$

which is undone by $v[2k] = m_k + d_k$ and $v[2k{+}1] = m_k - d_k$: a reversible relabeling that loses nothing $(24 + 24 = 48)$, applied independently to the horizontal and the vertical offsets.

The *symmetric* pattern v^{s} keeps only the common parts, setting both quadrupoles of every cell to m_k (all $d_k = 0$); the *antisymmetric* pattern v^{a} keeps only the differences, setting the two quadrupoles of each cell to $+d_k$ and $-d_k$ (all $m_k = 0$). Adding them back returns the original cell by cell — at the QF index $m_k + d_k = v[2k]$, at the QD index $m_k - d_k = v[2k{+}1]$ — so every misalignment is the sum of exactly one of each, $v = v^{\mathrm{s}} + v^{\mathrm{a}}$. The maps $v \mapsto v^{\mathrm{s}}$ and $v \mapsto v^{\mathrm{a}}$ are complementary projectors P_{sym} and P_{anti} with $P_{\mathrm{sym}} + P_{\mathrm{anti}} = I$; the two pattern families are orthogonal and each spans half of the 48-dimensional misalignment space, so *any* pattern resolves into one symmetric and one antisymmetric part. Applied separately to the two planes this gives $v_x = v_x^{\mathrm{s}} + v_x^{\mathrm{a}}$ and $v_y = v_y^{\mathrm{s}} + v_y^{\mathrm{a}}$, the representation used in the channel decomposition of Sec. III B.

The physical distinction between the two components follows from the opposite-sign gradients of QF and QD. A misaligned quadrupole deflects the beam by an angle proportional to (gradient)×(offset). When two neighbors are displaced *oppositely* (antisymmetric component), the opposite gradients and opposite offsets combine to kicks of the *same* sign: the kicks add coherently around the ring into a slowly varying deflection pattern whose dominant azimuthal harmonics lie near the betatron tune, and the closed-orbit response is large. When two neighbors are displaced *together* (symmetric component), the kicks are opposite and cancel within each cell; what survives is a cell-to-cell alternation at azimuthal harmonic $k \approx 24$ (one sign change per cell, 24 cells). The closed-orbit response to a deflection pattern of harmonic k follows the gain law

$$G_k = \frac{C}{|Q^2 - k^2|}, \qquad C \approx 24.8,\ Q^2 \approx 5, \tag{5}$$

which is resonant for k near the tune and strongly suppressed far above it; with $Q \approx 2.3 \ll 24$ the alternating deflection pattern of a symmetric component is nearly invisible [Fig. 1]. Quantitatively, a 10-μm-rms random pattern leaves an rms orbit trace of ~ 100 μm if antisymmetric but ~ 19 μm if symmetric (tracking-derived response), and the two traces differ in kind, not only in size: the symmetric trace is almost entirely *leakage*, a few-percent admixture of the pattern into the strongly-coupled modes. Correcting the orbit therefore cancels most of what an antisymmetric pattern produces but reaches only that few-percent leakage of a symmetric one — the symmetric misalignment itself is left nearly untouched, and with it the residual false EDM.

The same conclusion can be read off the response matrix itself without reference to harmonics. The singular-value decomposition of R supplies 48 mutually orthogonal, unit-normalized misalignment patterns v_i, each with its own orbit strength σ_i (pattern v_i produces an orbit of size σ_i). We ask how symmetric each of these patterns is by applying to it the very same symmetric projector P_{sym} defined above: the *symmetric fraction*

$$s_i = \|P_{\mathrm{sym}}\, v_i\|^2 \in [0, 1] \tag{6}$$

is the squared length of the pattern's projection onto the symmetric subspace, so that $s_i = 0$ for a purely antisymmetric pattern and $s_i = 1$ for a purely symmetric one. Equation (6) thus turns the decomposition of Eq. (4) into a single number per SVD mode that can be plotted against its orbit strength. Plotting σ_i against s_i [Fig. 1, right] shows the strongly-coupled patterns to be almost purely antisymmetric ($s_i \approx 0$) and the weakly-coupled ones almost purely symmetric ($s_i \approx 1$); the overall visibility ratio between the strongest and weakest patterns is $\mathrm{cond}(R) = 193$. Orbit correction, which by construction removes what the orbit sees, therefore removes the antisymmetric component and leaves the symmetric one.

B. The false EDM as a bilinear form: four channels

The connection between this decomposition and the false EDM follows directly from the physics of Sec. I A, with no further formalism. A horizontally misaligned quadrupole precesses the spin about the vertical axis by a small angle proportional to its offset $v_{x,i}$; a vertically misaligned one precesses it about the radial axis by an angle proportional to $v_{y,j}$. Two precessions about different axes do not commute, and the out-of-plane residue left by each ordered pair of them is proportional to the

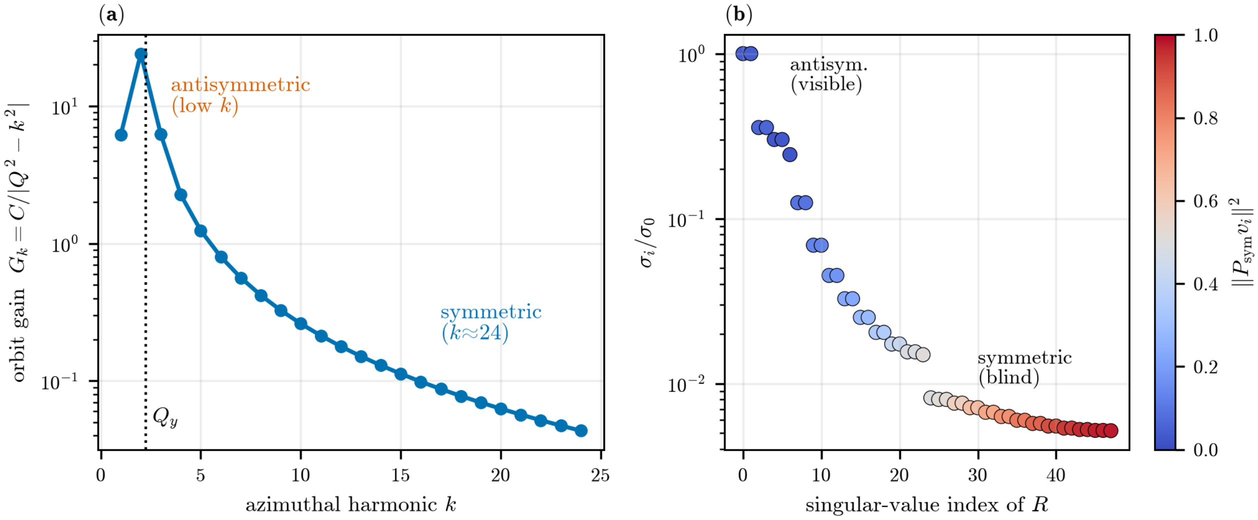

FIG. 1. Orbit visibility of misalignment patterns (analytic lattice, $Q_y = 2.30$). Left: the orbit gain law of Eq. (5); antisymmetric patterns drive low harmonics near the betatron resonance, symmetric patterns drive $k \approx 24$, far above it. Right: the independent misalignment patterns ranked by the strength of their orbit response; patterns that couple strongly are antisymmetric (QF, QD opposite), those that couple weakly are symmetric (QF, QD together); the visibility ratio is cond$(R) = 193$.

product of the two small angles. Summing this over all pairs of quadrupoles around the ring,

$$f \;=\; \sum_i \sum_j W_{ij}\; v_{x,i}\, v_{y,j} \;\equiv\; B(v_x, v_y), \tag{7}$$

where the pairwise weights W_{ij} depend only on the lattice — on where the two magnets sit and on how the orbit deflection started by one is carried around the ring to the other [the orbit response, Eq. (3)] — and not on the misalignments themselves. Equation (7) states that the false EDM is *bilinear*: every term contains exactly one horizontal and one vertical offset, never two of the same plane. This is the sign structure verified directly in Sec. II B.

Splitting each plane independently into its symmetric and antisymmetric parts, $v_x = v_x^{\rm s} + v_x^{\rm a}$ and $v_y = v_y^{\rm s} + v_y^{\rm a}$ [Eq. (4)], and expanding the double sum, the terms fall into four groups:

$$f = \underbrace{B(v_x^{\rm s}, v_y^{\rm s})}_{f_{ss}} + \underbrace{B(v_x^{\rm s}, v_y^{\rm a})}_{f_{sa}} + \underbrace{B(v_x^{\rm a}, v_y^{\rm s})}_{f_{as}} + \underbrace{B(v_x^{\rm a}, v_y^{\rm a})}_{f_{aa}}. \tag{8}$$

The two planes are mixed by the weights W_{ij}, not by the projectors. Because a misalignment component acts on the spin mainly through the orbit it drives, and that orbit is large for an antisymmetric component and G_k-suppressed for a symmetric one [Eq. (5)], the pure-symmetric channel f_{ss} is by far the smallest, and every channel containing an antisymmetric factor is larger.

We stress that f_{sa} and f_{as} are two distinct physical quantities, not a double counting. In f_{sa} the *horizontal* pattern is symmetric and the *vertical* one antisymmetric; in f_{as} the roles are exchanged. Nothing forces these to be equal: the two planes enter the ring differently (only the horizontal plane carries the bending), and the weights W_{ij} are not symmetric under exchanging the planes, because the order of two non-commuting precessions matters. The tracker confirms the difference directly: for the representative pattern, $f_{as} = -272\times$ target while $f_{sa} = +28\times$ — an order of magnitude apart and of opposite sign.

Each channel is a separate tracking measurement, not a fit, and this measurement underlies all symmetric/antisymmetric attributions in this paper. The misalignment pattern (v_x, v_y) is projected once with Eq. (4), and the tracker is then run four times. In each run the quadrupole offsets of the machine are *set to* one projected part per plane — for f_{as}, for example, the horizontal offsets are the pattern $v_x^{\rm a}$ and the vertical offsets are $v_y^{\rm s}$, with no other imperfections — and the estimator of Sec. II B returns that channel's f directly. The weights W_{ij} never need to be known or computed: they are properties of the lattice that the tracker itself embodies, and Eq. (7) enters only through its structural consequence that every term contains exactly one horizontal and one vertical offset, so that the four runs isolate the four groups of terms. Bilinearity is then checked rather than assumed: Figure 2 shows the result at $\sigma = 10\ \mu$m: the four signed channels sum to the full-pattern false EDM within 0.1% ($f/\sum = 0.999$–1.001 over three seeds), confirming Eq. (8); the pure-symmetric channel is the smallest in every seed; and each channel individually scales as σ^2. The measured channel ratios are consistent with the bilinear structure: one orbit-trace factor of ~ 5 enters per plane, so f_{aa}/f_{ss} of order tens is expected and observed. The measured channels are used as the quantitative reference throughout.

Two consequences frame the rest of the paper. First, the false EDM is not a function of the rms misalignment: it is σ^2-homogeneous, but at fixed σ its value depends on the pattern — in Fig. 2 the channel magnitudes at the same 10 μm rms range from $\sim$20× to $\sim$400× the target in the three-seed means, and from $\sim$1× to $\sim$900× across individual seeds. A single alignment tolerance therefore cannot summarize the systematic, and a correction procedure changes not only the size but the *statistics* of the misalignment: it converts a random pattern into a structured one, concentrated in whichever component the procedure cannot see. Second, the division of labor for suppression is fixed by Eq. (8): orbit correction removes the antisymmetric factors and with them the three large channels; the remaining floor is f_{ss}, and any further progress requires a measurement that reaches the symmetric component itself.

IV. THE REFERENCE SOLUTION AND ITS COST

The design study of Ref. [1] controls the symmetric misalignment with two ingredients: beam-based alignment (BBA), which reaches the symmetric component that orbit inversion cannot, and a four-fold polarity measurement that cancels the residual false EDM. We reproduce both in tracking and then quantify what each costs. The picture that emerges — BBA reaches the target but only after many passes, and the four-fold reaches it at once yet only if the machine is held identical across a sequential polarity reversal — is what motivates the orbit-only alternative of Sec. V.

A. Procedure

Standard quadrupole beam-based alignment does not invert a matrix and does not read an amplitude against a calibration; it locates each magnetic center as the beam position at which the response to a gradient change *vanishes*. For each of the 47 alignable quadrupoles in turn:[2] (i) using the nearest steering corrector, scaled by the single response-matrix element R_{ij}, the beam is placed at two positions bracketing the expected center of quadrupole i; (ii) at each position the gradient of quadrupole i is changed by a small amount and the difference of the two closed orbits is recorded at all 48 BPMs — this difference is the feed-down imprint, proportional to the beam–center offset; (iii) since the imprint changes sign as the beam crosses the center, the center is obtained as the weighted least-squares zero over the BPMs, with the spread of the per-BPM zero crossings retained as a per-magnet quality measure. All orbits are read with the beam placed on the 4D closed orbit (Sec. II B). In the simulation the steering corrector is a controlled transverse displacement of a quadrupole adjacent to the one under test; in the experiment the same dipole kick can be produced without moving any magnet, by a current asymmetry between the quadrupole coils, which shifts the magnetic center electrically. The two are equivalent — an offset quadrupole and an on-axis quadrupole with a superimposed dipole field act identically on both orbit and spin.

The role of the response matrix in this procedure is confined to the *forward* direction: a single, well-conditioned element sizes the steering bump. No global inverse appears anywhere, so the conditioning wall of Sec. V B does not apply; the 47 centers are measured locally and independently, and the symmetric combination of the answers is determined as well as any other combination. An error in the forward element mis-sizes a bump and biases one local measurement — an effect that will be visible below — but it does not re-introduce the collective visibility gap.

B. Idealized and realistic optics

In idealized optics the procedure works outright: the symmetric centers are recovered, and the false EDM of the representative pattern drops from 356× to 1.6× the target.

Under a conservative 1%-gradient-error optics ($\approx 5\%$ β-beat, above the 1–2% LOCO range and thus a stress test) a single pass fails, through a mechanism we call optics *breathing*: changing one quadrupole's gradient to find its center slightly shifts the tune and focusing functions of the *whole* ring, which re-transports the large pre-existing closed orbit (built by all the other misaligned quadrupoles) and moves it at every BPM by an amount that scales with that orbit. This spurious motion, coherent with the modulation and hence un-averageable, biases the zero crossing. (The same breathing defeats the amplitude-readout estimators of Sec. V B, quantified there.) The remedy is iteration — correct the orbit with the current estimates, then re-measure — since a smaller orbit breathes less. Iterating with under-relaxation and the per-magnet quality cut, the vertical plane converges cleanly (residual roughly halving each pass, reaching 0.06 μm by the eighth). The horizontal plane initially stalled, and the diagnosis is instructive. The steering bumps were sized with the analytic optics model, which treats the deflectors as drifts (Sec. II A). Vertically this is accurate, and indeed the model reproduces the tracker's vertical tune to four decimal places (fractional tune 0.3032 in both). Horizontally the deflectors focus and bend, and the model does not know it: it predicts equal tunes in the two planes (2.30), while the tracker's horizontal tune is 2.64, higher by ~ 0.34; ele-

[2] The gradient-modulated quadrupole of cell 0 is a distinct element type in our model and is excluded; its offset is set to zero and reported as such.

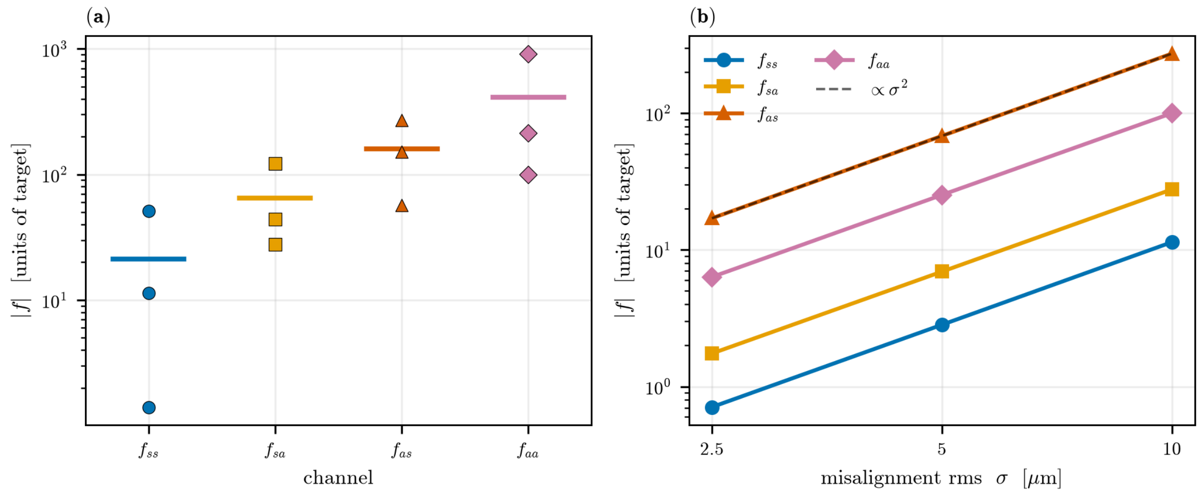

FIG. 2. The four false-EDM channels of Eq. (8), measured by projecting the misalignments and tracking each part separately. (a) At $\sigma = 10\ \mu$m the pure-symmetric channel f_{ss} is the smallest, and every channel containing an antisymmetric factor is larger (markers: individual seeds; horizontal line: their mean; the spread reflects the signed, bilinear character of f). (b) Each channel scales as σ^2. The four signed channels sum to the full signal within 0.1%.

ment by element, the model's horizontal response matrix is essentially uncorrelated with the measured one (median ratio -0.05), and it selects the wrong steering corrector for 36 of the 47 magnets, against a 5%-accurate response and 12 wrong selections vertically. Re-building the steering from the *measured* response matrix — ordinary practice on a real machine — repairs the horizontal plane: the per-magnet quality ceases to throttle the horizontal corrections, and the false EDM after three passes improves fivefold over the analytic-model steering ($28\times$ versus $145\times$); iterated with the measured matrix it falls monotonically to sub-target by the eighth pass (Fig. 7), while the analytic steering stalls near $145\times$. The measured matrix is obtained in the simulation exactly as it would be on the machine: each quadrupole in turn is displaced by a known amount, the resulting closed-orbit change is recorded at all 48 BPMs, and the ratio (orbit change)/(displacement) fills one column of the matrix.

C. The polarity four-fold and its cost

Reference [1] reaches the symmetric channel by a route that does not pass through any orbit reading: *reversing the polarity of all focusing quadrupoles* and repeating the counter-rotating difference [its Eq. (C2)]. Under simultaneous reversal of the beam direction and the magnetic gradient the Lorentz force $q\,\boldsymbol{v} \times \boldsymbol{B}$ is unchanged, so a counter-rotating beam in the reversed lattice retraces the trajectory of a co-rotating beam in the nominal one; the magnetic-lattice false EDM is identical between the two configurations and cancels in the four-fold combination, while the electric-field EDM does not. We confirm this directly: the four-fold combination removes the false EDM of *both* the symmetric and the antisymmetric patterns to the estimator floor ($0.009\times$ and $-0.017\times$ the target), independently of quadrupole roll or β-beating, which respect the same $\boldsymbol{v} \times \boldsymbol{B}$ symmetry.

The four-fold cancellation is exact in principle, and it reaches the symmetric channel that BBA alone leaves above target — but it is not free. The two polarity states cannot be stored at once; they are measured in sequence, and the cancellation requires the machine to present the *same* misalignment in both. Because the false EDM is bilinear [Eq. (7)], the residual left by any change between the states is proportional to the *product* of the ambient misalignment and the change, so the same drift costs less on a better-aligned machine: after BBA (symmetric rms $\sim 2\ \mu$m) a $1\ \mu$m symmetric inter-configuration drift leaves $0.07\times$ the target, whereas on the raw $10\ \mu$m machine it produces tens of times the target. More generally a fractional error ε in reproducing the reversed gradients leaks a fraction of the large common-mode signal into the EDM channel; an $\varepsilon = 10^{-3}$ gradient asymmetry leaves about $1\times$ the target on a $10\ \mu$m machine. The drift that matters is the symmetric one, invisible to the orbit, so the four-fold demands a BBA-level symmetric alignment *held stable* across the reversal. This — the alignment campaigns, the sequential second configuration, and the stability requirement between them — is the operational cost the orbit-only method of the next section removes.

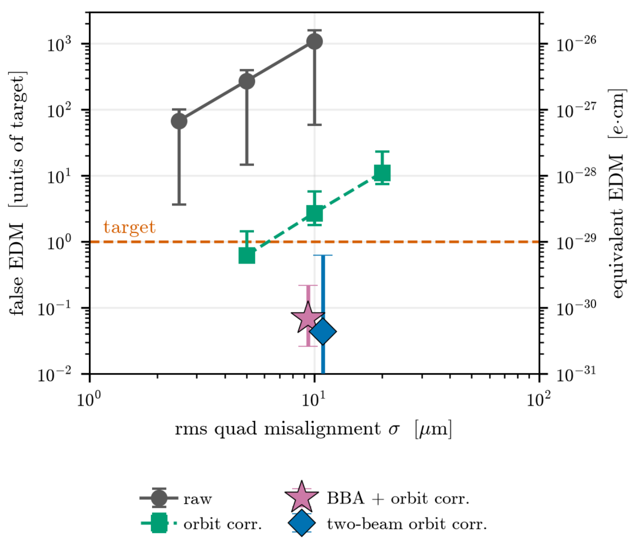

FIG. 3. The single-beam false EDM versus rms misalignment (tracking; median and full spread). Circles: the raw pattern (no correction). Squares: orbit correction alone, which removes the antisymmetric channels but leaves the symmetric floor — still a few times the target at $\sigma = 10\ \mu$m. Both scale as σ^2 ($p = 2.00$ at each stage), confirming the geometric-phase origin; the fit verifies the scaling, not an absolute floor, since at fixed σ the false EDM varies by more than an order of magnitude with the symmetric/antisymmetric content of the pattern (Sec. III B). The symmetric floor is reached below target by two independent routes at $\sigma = 10\ \mu$m (fixed rcond $= 0.01$). Star: beam-based alignment followed by one orbit correction (median 0.07×, worst 0.22×, five seeds; Table II). Diamond: two-beam counter-rotating orbit correction against a calibrated reference, no BBA, at a realistic 1% β-beat (median 0.043×, worst 0.62×, fifteen seeds; Table I and Sec. V C). The diamond is the odd remnant $\frac{1}{2}|f_{\rm CW} - f_{\rm CCW}|$; both routes place the residual well below the target.

V. REACHING THE SYMMETRIC COMPONENT WITH COUNTER-ROTATING ORBIT CORRECTION

A. The residual after standard correction

For random misalignments of $\sigma = 10\ \mu$m in both planes, the raw false EDM is of order 10^3 times the target [Eq. (1)]; a representative pattern used throughout this paper gives 356×. Standard orbit correction — flattening the closed orbit with steering correctors — reduces the ensemble median by a factor of about eight; combined with the counter-rotating–beam difference discussed in Sec. V C, the residual is about 47× the target ($\approx 5\times10^{-28}$ $e$cm), scaling as σ^2 (Fig. 3). Orbit correction is thus effective but not sufficient; the residual is the symmetric floor of Sec. III, and the rest of this section shows why a single-beam correction cannot reach it and how correcting the two counter-rotating beams jointly does.

B. The symmetric component cannot be recovered by inversion or amplitude readout

Before presenting the measurement that succeeds, we summarize the techniques that do not, and why. All of them attempt to reconstruct the misalignments (including, crucially, their symmetric part) from orbit readings. The realistic error model includes static BPM offsets ($\sim 100\ \mu$m each, the electronic zero of a monitor relative to the true axis), white BPM noise ($\sim 1\ \mu$m per reading, reducible by averaging), and a coherent optics-model error (β-*beating*, implemented in the tracker as per-quadrupole fractional gradient errors; the 1% rms operating point used in the tests below produces $\approx 5\%$ rms β-beating (peak $\approx 10\%$) in this lattice, computed from the perturbed one-turn Twiss, so it is if anything pessimistic relative to the $\sim$1–2% typical of LOCO-corrected optics).

Difference-orbit inversion. Modulating all quadrupole gradients between two settings and subtracting the orbits cancels the static BPM offsets exactly and leaves a linear system $\Delta y = \Delta R\, \Delta q$. With no errors the inversion is exact. Its weakness is conditioning: $\mathrm{cond}(\Delta R) = 3.7\times10^4$, with the weak directions being precisely the symmetric ones (established by the same symmetric-fraction analysis as in Fig. 1, applied to ΔR), so reconstruction errors are amplified along the component of interest. Regularizing the inversion — truncated SVD or Tikhonov damping — does not help here: truncation discards precisely the symmetric directions, trading their noise amplification for a full bias of the sought component. This much is already true *without* any optics error: at the lock-in noise floor alone the conditioning amplifies the symmetric reconstruction error to about ten times the antisymmetric one (Fig. 4, at zero β-beat). White noise can be beaten by synchronous detection — modulating at ~ 1 kHz and averaging reaches an effective noise of $\sim$14 nm within minutes — but a coherent optics-model error cannot: it is identical in every average, and already at the gentlest realistic β-beating — 1% (0.2% rms gradient error), the best a LOCO correction achieves — the reconstructed symmetric component is several times the true one ($\sim$110 μm against a 41-μm signal in our standard test) while the antisymmetric component remains accurate [Fig. 4]; the error grows with the β-beat and is larger still at the 5% operating point (1% rms gradient error) adopted for the constructive tests of Secs. IV–V C, so the symmetric channel is lost across the whole realistic range. Because the limit is coherent, it is independent of BPM technology; lower-noise pickups, including SQUID-based ones, shorten the integration time but do not move the floor. The same structural point can be stated as a bound: any estimator that cancels static offsets by a two-setting difference pays for the cancellation with a $\|\Delta R^{-1}\| \sim \|R^{-1}\|/\varepsilon_g$ amplification, where ε_g is the relative setting difference.

Per-quadrupole modulation read as an amplitude. Assigning each quadrupole its own modulation frequency

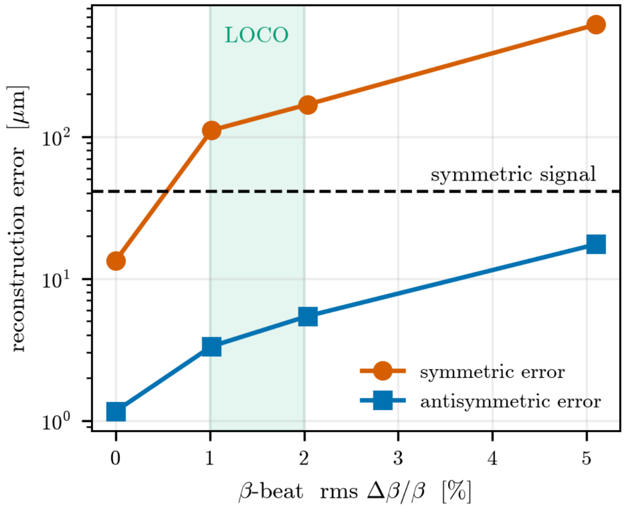

FIG. 4. Difference-orbit inversion at the synchronous-detection noise floor (10 nm, 40 seeds): reconstruction error of the symmetric and antisymmetric components versus the β-beat (the driving gradient error σ_g is $\approx 5\times$ smaller). The dashed line marks the size of the symmetric signal itself, and the shaded band the realistic LOCO range ($\sim$1–2% β-beat). Already at 1% the symmetric error exceeds the signal; even at zero β-beat the conditioning leaves the symmetric error an order of magnitude above the antisymmetric one, though the Pearson correlation between input and reconstruction stays misleadingly high.

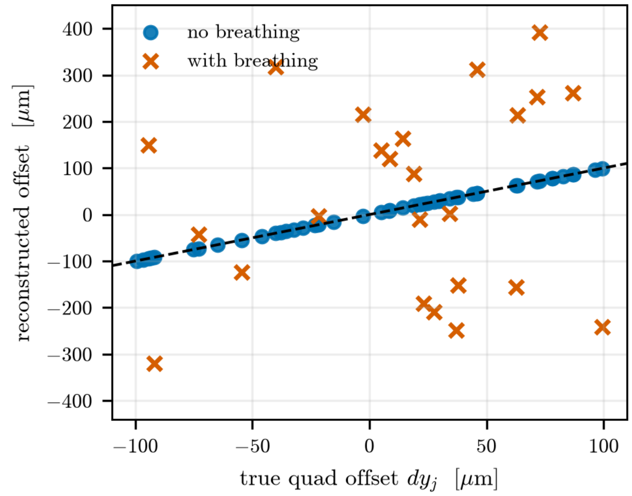

FIG. 5. Per-quadrupole gradient modulation read as an amplitude: reconstructed versus true offsets. The reconstruction is exact when the optics breathing is artificially removed, but uncorrelated with it present — the breathing, coherent with the modulation, does not average down.

and reading the per-frequency amplitude at the BPMs — the ac-BBA concept of Ref. [5] — promises a direct, inversion-free readout: a quadrupole whose center is offset from the beam by o_i imprints, when its gradient is modulated, a dipole-like deflection proportional to o_i (the *feed-down* kick, the dipole field component seen by a beam passing off-center). The method fails in this ring for an instructive reason [Fig. 5]. The problem is caused by the modulated quadrupole itself, acting globally. A quadrupole is a focusing element: changing its gradient by even 2% slightly shifts the tune and the focusing functions of the *entire* ring. The pre-existing closed orbit — built by the offsets of all the other quadrupoles, and at $\sim$ 0.37 mm rms several hundred times larger than the $\sim$ 0.9-μm feed-down signal being sought — is re-transported through this slightly altered focusing and moves by a few μm at every BPM. This global response — the optics *breathing* already met in Sec. IV B — appears at exactly the modulation frequency of the quadrupole being measured, so no frequency filter can separate it from the signal. The reading one then demodulates at a BPM is therefore not the quad's own feed-down: for the worst-coupled quadrupole the amplitude driven by the breathing of the pre-existing orbit is 266 times larger than the amplitude its own offset would produce, so the number one would invert for the offset is dominated by the orbit of the *other* magnets, and within this single-frequency readout no fixed calibration recovers the right quantity from it [Fig. 5]. Because breathing is coherent with the modulation it does not average away, and removing it in an idealized model restores perfect reconstruction, which pins the failure entirely on this mechanism.

Learned and structured estimators do not escape the limit either: because the misalignment-to-orbit map is linear, a trained neural network, sparsity priors (LASSO, CLEAN), and a calibrate-then-monitor strategy (LOCO) all pass through the same response matrix and inherit its blindness to the symmetric component — in our tests a neural network matched truncated-SVD inversion to within its noise (5.6 vs. 6.3 μm symmetric error against a 9.9 μm signal), confirming that the limit is the information in the data, not the flexibility of the estimator.

Counter-rotating beam separation and the tune. The control chain of Ref. [1] monitors the *separation* of the two counter-rotating (CR) closed orbits and reduces it with correctors. The CR separation, however, is itself an orbit observable and inherits the orbit's visibility structure: in our tracking the antisymmetric-to-symmetric signature ratio is 5.7× in a single-beam orbit and 8.0× in the CR separation (an independent pattern set gives 3.8× and 4.5×) — the CR separation opens no additional window on the symmetric component. The separation, however, is the *difference* of the two counter-rotating orbits; that their *sum* does open such a window — and with it a joint correction that reaches the symmetric component — is the subject of Sec. V C below, and is the central result of this work. The polarity reversal of Ref. [1] reaches the symmetric component by a different route again, cancelling the geometric phase directly rather than through any orbit reading, at the cost analyzed in Sec. IV C. Finally, raising the betatron tune toward the symmetric kick harmonic, which would make that component visible in principle, is capped by the stopband ($Q \leq 12$, against the $Q \geq 16$ the slowly-varying symmetric modes require) and in any case raises the false EDM itself steeply (a measured $\sim g^3$).

The common conclusion of this section is structural. For every technique in the inversion and amplitude-readout families, the limiting effect is coherent — conditioning amplified by β-beating, or optics breathing — and coherent effects do not average down. *Within the simulated model, no BPM technology — including SQUID-based BPMs — changes these floors.* We emphasize the scope of this statement. It is a structural property of the estimator classes considered, established by class-level arguments (the conditioning bound and the coherence of the breathing term) together with systematic simulation; it is not an information-theoretic impossibility claim for all conceivable orbit-based measurements. Indeed, the counter-rotating correction of the next subsection is an orbit-based measurement outside these classes that does reach the symmetric component.

C. Correcting both counter-rotating orbits

The measurement of Ref. [1] that isolates the genuine EDM is the counter-rotating difference: the EDM is odd under beam reversal, so it survives $(dS_y/dt)_{\rm CW} - (dS_y/dt)_{\rm CCW}$ while common-mode systematics cancel [Eq. (C1)], and two beams stored at once make the difference drift-free. A false EDM that were purely beam-reversal-even would vanish here; the geometric phase does not, because the two beams ride slightly different closed orbits, and a beam-reversal-*odd* remnant survives. At $\sigma = 10$ μm over five seeds this remnant is 4–60× the target (median 22×) for a purely symmetric misalignment and 38–496× (median 175×) for a purely antisymmetric one; correcting a *single* beam's orbit removes the antisymmetric part but leaves the difference at $\sim 45\times$ (median, 0.1–67×; Table I).

The reason a single correction is not enough, and the way past it, are both contained in the relation between the two orbit response matrices. They are nearly opposite: writing them for a horizontal quadrupole displacement, $R_{\rm CW} = -R_{\rm CCW} + \Delta$, the sign-reversed part dominates and a direction-common residue Δ survives. The sign flip is the leading effect because a displaced quadrupole acts as a fixed dipole (its feed-down field is set by the displacement, not the beam), and the Lorentz force $q\,\boldsymbol{v} \times \boldsymbol{B}$ reverses with the velocity of the counter-rotating beam, so the same kick deflects the two beams oppositely. What it does *not* reverse is the betatron phase advance between magnets, which is not mirror-symmetric about each element in a ring with a definite QF/QD/deflector sequence; that non-reciprocity is the residue Δ. In Frobenius norm $\|\frac{1}{2}(R_{\rm CW} - R_{\rm CCW})\| = 33.8$ and $\|\frac{1}{2}(R_{\rm CW} + R_{\rm CCW})\| = 7.3$ for the horizontal plane, a Δ of 21%; the vertical plane is the same (45.8 and 9.7, 21%), so the effect is not an artifact of the horizontal bending. This near-opposite- ness is exactly why the *separation* of the two orbits — the observable the reference control chain monitors (Sec. V B) — is blind to the symmetric component: the difference $R_{\rm CW} - R_{\rm CCW} \approx -2R_{\rm CCW}$ doubles the antisymmetric-visible part, and its symmetric visibility, the Frobenius-norm ratio $\|R\,P_{\rm sym}\|/\|R\,P_{\rm anti}\|$ of the response of unit-normalized symmetric versus antisymmetric patterns, is 0.19 — no better than a single beam (0.24). The *sum*, by contrast, cancels the dominant antisymmetric response and lays the symmetric component bare: its symmetric visibility is 1.01 and its condition number is 30. The information the single beam and the separation both lack is carried in the sum of the two counter-rotating orbits.

Figure 6 shows this in real space. For a representative antisymmetric misalignment the two beams ride near-mirror-image closed orbits (correlation −0.95 at zero azimuthal shift) whose large amplitudes cancel in the sum, leaving a small residue. For a symmetric misalignment the two orbits are instead the *same* shape displaced by a few BPMs — their correlation is near zero at zero shift but rises to +0.94 at a $\sim$5-BPM shift, a consequence of the direction-reversed betatron phase advance that constitutes the residue Δ — so the sum does not cancel and survives at the scale of the original symmetric orbit. The sum thus suppresses precisely the antisymmetric orbit that masks the symmetric one in a single beam, and preserves the symmetric orbit that a single beam cannot see.

Correcting both orbits jointly uses it. In the frozen-spin ring the clockwise and counter-clockwise beams are stored simultaneously and counter-propagate through each other; with many bunches the beam-position monitors resolve the two species in separate time windows, so the per-BPM sum of the two closed orbits is directly available. Stacking the two response matrices, $S = [R_{\rm CCW}; R_{\rm CW}]$, and solving the single least-squares correction that flattens both orbits is equivalent to using their sum and difference together; it therefore has access to the symmetric component through the sum. The effect is a conditioning one, made sharp by the regularization. The single-beam horizontal matrix (from tracking) has cond$(R_{\rm CCW}) = 135$ (228 vertically); at the truncation of the pseudo-inverse (rcond $= 10^{-2}$) seventeen of its weakest singular directions fall below the threshold and are discarded, and every one is symmetric-dominated — retaining them would amplify the BPM noise on the symmetric correction by $\sim 7\times$, the same obstruction as the inversion no-go of Sec. V B. The stacked matrix has cond$(S) = 73$ (118 vertically): no direction falls below the threshold, and the symmetric modes are corrected with a noise amplification of 0.7×. The second beam does not circumvent the no-go; it lifts the symmetric component above the noise floor the no-go is about, by supplying the sum information no single beam carries.

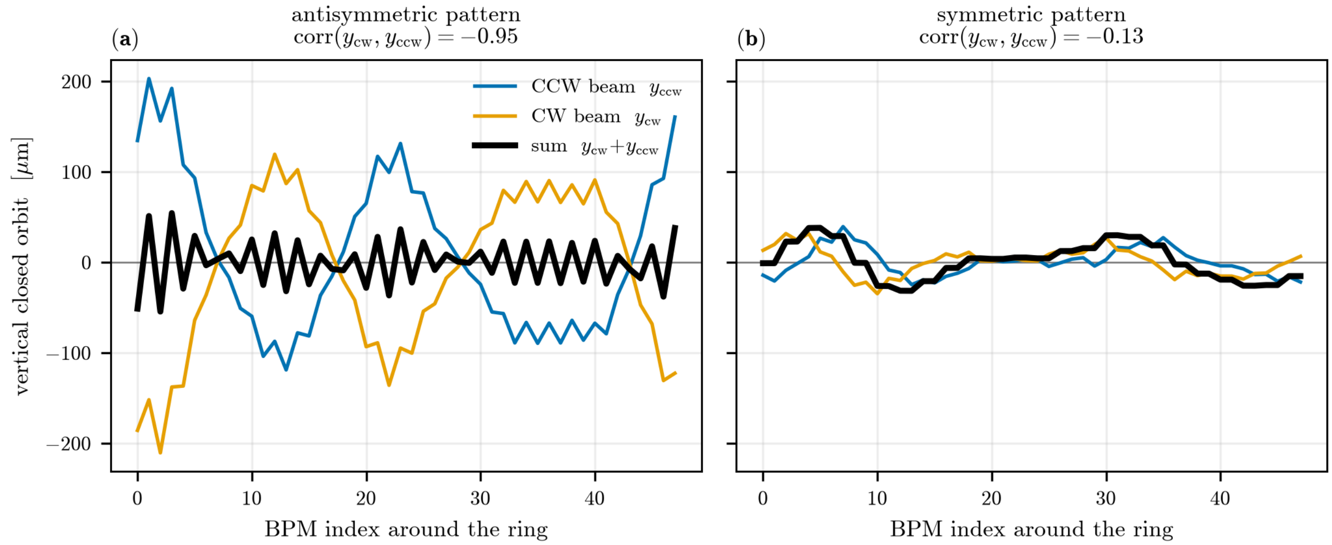

FIG. 6. Vertical closed orbits of the two counter-rotating beams for the same 10 μm-rms misalignment (analytic response matrices). (a) An antisymmetric pattern drives large, near-opposite orbits (corr = −0.95) that cancel in the sum (black). (b) A symmetric pattern drives small orbits that are the same shifted by a few BPMs (corr ≈ 0 at zero shift); their sum does not cancel and stays at the scale of the individual orbits. The sum therefore removes the antisymmetric orbit that swamps a single beam and keeps the symmetric orbit that a single beam cannot resolve.

What performs the correction matters, and is fixed by the same physics. The quantity solved for is a set of *magnetic-center shifts* c — one per alignable quadrupole, 47 per plane — realized, as in the BBA of Sec. IV A, by a current asymmetry between the quadrupole coils, not by dipole steering. The distinction is essential. A center shift is a geometric change of the misalignment itself, $m \to m - c$, the *same* for both beams, so it can null the symmetric component at its source; a dipole steering corrector, by contrast, displaces the beam rather than the quadrupole, and because the Lorentz kick reverses with the beam velocity its effect on the beam-to-center offset is direction-odd — it can flatten one beam's orbit but cannot center both beams in the quadrupoles at once. Two-beam correction therefore presumes center-shift actuators, the hardware the alignment program already requires; it introduces no new class of corrector, only a new use of the two orbits.

The consequence for the false EDM is decisive. On the raw 10 μm machine, with no beam-based alignment and the orbit measured against a calibrated reference, two-beam orbit correction reduces the symmetric misalignment residue from $\sim$5.5 μm (single beam) to 0.1–0.7 μm — at or below the $\sim$2 μm that BBA reaches (Sec. IV) — and drives both counter-rotating false EDMs to the floor together, so the odd difference falls below target.[3] Table I gives the dependence on the coherent optics error. Through the entire realistic range the correction is robustly sub-target: at the typical LOCO operating point (1% β-beat, 0.2% rms gradient error) the odd remnant is a median 0.043× the target over fifteen seeds, all sub-target (worst 0.62×), and on a well-corrected machine (0.5% β-beat) a median 0.03×. Only at an unrealistically large 5% β-beat, above what LOCO delivers, does the coherent residue begin to matter (median 0.66×, three of five sub-target) — the same coherent-model-error floor that defeats the difference-orbit inversion of Sec. V B, here softened because two-beam correction is a well-conditioned forward flattening rather than an inversion. Adding a 0.2 mrad unmodeled quadrupole roll to the 1%-β-beat machine leaves the result unchanged (median 0.12×): roll is a magnetic-lattice effect, even under the $\boldsymbol{v} \times \boldsymbol{B}$ symmetry, so it contributes equally to the two beams and cancels in the odd difference.[4]

These figures presuppose a calibrated reference, and the qualification is not a technicality. The static electrical offsets of real BPMs — $\sim$100 μm, the same offsets the reference control chain removes by differencing (Sec. V B) — degrade this scheme for a sharp and instruc-

[3] The scheme reads the two counter-rotating orbits on the shared BPMs, as the reference design already does to monitor their separation, distinguished by bunch timing; unlike that separation it uses the two absolute orbits, not their difference. The 14 nm figure is the effective per-point noise reached by synchronous averaging over minutes (Sec. V B), and enters only as the white-noise floor of the correction.

[4] The roll here enters the false-EDM measurement, its dominant geometric-phase channel; its second-order coupling of the two transverse orbits is not fed into the (uncoupled) correction, so the test bounds the leading channel that the $\boldsymbol{v} \times \boldsymbol{B}$ argument identifies.

tive reason. The offset b is common to the two beams, so it cancels in their difference but adds *doubly* to their sum — and the sum is precisely the channel that carries the symmetric component. Symmetric misalignment and static offset are therefore *degenerate* in the counter-rotating sum: the very projection that exposes the one exposes the other, and no single-time orbit measurement separates them. In tracking, restoring 100 μm uncalibrated offsets returns the odd remnant from 0.043× to $\sim$200× the target (Table I), and the tolerance is severe — the symmetric residue grows as 0.96× the offset rms, so sub-target operation would demand the offsets known to $\lesssim$0.5 μm, the quadrupole-center level that only BBA provides. Two-beam correction thus does not remove the need for an absolute position reference; it concentrates that need into a single, cleanly stated degeneracy.

What it *does* remove is the *periodic* realignment. Drift is differential: between two orbit acquisitions the static offset is unchanged, so it cancels identically in the orbit *change* $\Delta y = y(t) - y(t_0) = R\,\Delta q$, and the counter-rotating sum of Δy exposes the symmetric part of the *drift* with no offset contamination whatever (in tracking the leaked symmetric residue is 0.000 μm against 100 μm offsets). A continuous two-beam correction acting on the differential orbit therefore holds the symmetric drift at the offset-free level of Table I, and — β-beat and roll being harmless — robustly so. This fixes the role of counter-rotating orbit correction. It is not a substitute for the initial absolute alignment, which one BBA campaign, or the polarity four-fold of Sec. IV C, must still establish. But once that alignment is set, two-beam correction maintains it against drift — including the *symmetric* drift that a single-beam feedback cannot see, and that in the reference program forces the BBA campaign to be *repeated*. The operational consequence is developed in Sec. VI.

D. The alignment route: BBA closed by a final orbit correction

Counter-rotating correction is the orbit-only route to the symmetric floor. For completeness we record the alternative — the *alignment* route of Sec. IV — which reaches the same floor by pairing BBA with orbit correction, now that both tools are in hand. It also makes precise the sense in which each tool alone is limited.

The residual of the measured-matrix run falls monotonically once the steering is well conditioned: for the representative seed the false EDM is 28×, 21×, 10×, 4×, 1.0× the target at passes 3–7, and reaches 0.09× — below target — by the eighth pass (Fig. 7). There is no plateau. BBA lowers *both* the symmetric and the antisymmetric residual each pass, so with enough passes the false EDM crosses below the target on its own; the alignment route does not, in fact, require anything beyond BBA to reach the floor. What it does slowly, though, a single orbit correction does at once. At intermediate passes the residual

TABLE I. Counter-rotating orbit correction on the raw σ = 10 μm machine (no BBA; 14 nm BPM noise; tracking). The odd remnant is the counter-rotating difference $C = \frac{1}{2}|f_{\mathrm{CW}} - f_{\mathrm{CCW}}|$ in units of the target. Top block: correction against a calibrated reference, versus the coherent optics error — two-beam correction is sub-target across the realistic range and degrades only at an unrealistically large 5% β-beat. Bottom: with 100 μm uncalibrated static BPM offsets the odd remnant returns to $\sim$200×, because offset and symmetric misalignment are degenerate in the counter-rotating sum — unless the correction acts on the *differential* (drift) orbit, in which the static offset cancels identically. The LOCO operating point is over 15 seeds, the other points over 5.

β-beat (σ_g), reference	single-beam	two-beam	(median)
0.5% (0.1%), calibrated	$\sim$46×	0.03×	(5/5)
1% (0.2%, LOCO), calibrated	$\sim$32×	0.043×	(15/15)
1% +0.2 mrad roll, calibrated	$\sim$45×	0.12×	(5/5)
5% (1%), calibrated	$\sim$29×	0.66×	(3/5)
1%, 100 μm offset (absolute)	$\sim$650×	198×	(0/5)
1%, 100 μm offset (differential)	—	0.043×	(15/15)

is dominated by the antisymmetric part (the channel decomposition of the fifth-pass residual gives $f_{\mathrm{anti}} \approx 7.9\times$ against a negligible f_{sym}), and that part is orbit-*visible*; a single standard orbit correction applied to the BBA residual removes it collectively and reaches the target in about three passes rather than eight (median 0.07× over five seeds, Fig. 3), with a clean σ^2 dependence. The final orbit correction is thus an accelerator of the alignment route, not a precondition for it.

The resulting picture within the alignment route is consistent and complementary: *single-beam orbit correction removes the antisymmetric channels of Eq. (8) but leaves the symmetric floor (the counter-rotating remnant sits at 47×); null-finding BBA reaches the symmetric channel that single-beam orbit correction cannot see, and iterated far enough reaches the target on its own; combining the two reaches it in a few passes.* That the symmetric channel is not, in fact, beyond the reach of orbit correction altogether — that correcting the two counter-rotating beams jointly reaches it without BBA — is the subject of Sec. V. The division of labor is also one of *precision*: each per-magnet zero crossing carries a $\sim\mu$m-level measurement floor (residual breathing and steering-model error), so iterating BBA alone approaches that floor only slowly; the antisymmetric residue, by contrast, needs no per-magnet measurement at all — the orbit displays it directly and resonantly amplified, and nulling it collectively with a truncated-SVD correction (the "depth" above is the singular-value cutoff) is far more precise than measuring it magnet by magnet. Table II summarizes the chain. The endpoint is characterized over five misalignment seeds: at a fixed truncated-SVD depth (rcond = 0.01, chosen once and not tuned per seed) all five fall below the target, with median 0.07× and worst case 0.22× (Fig. 3). Because the false EDM is a

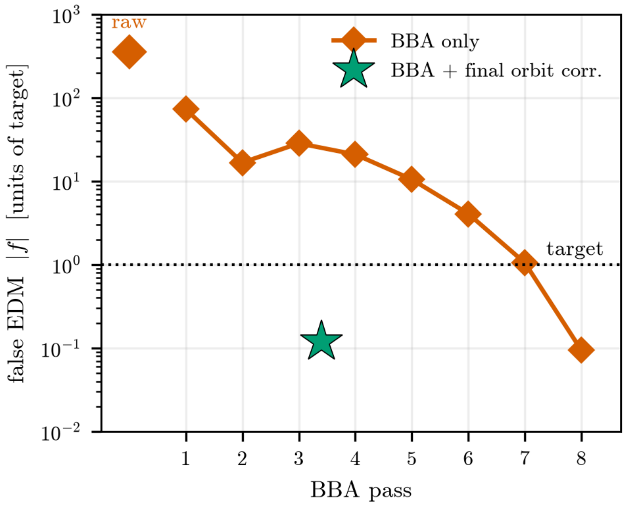

FIG. 7. Beam-based alignment drives the false EDM below target without a plateau (measured-matrix BBA under a conservative 1% rms gradient error, ≈ 5% β-beat, eight passes, C++ tracking). Diamonds: the false EDM $|f|$ from the raw machine (356× target) through eight BBA passes; it falls monotonically past the third pass and crosses below target by the eighth (0.09×). It does not plateau; the descent is merely slow, because BBA grinds the symmetric residual down magnet by magnet, reaching the orbit-blind component that a single-beam response-matrix inversion cannot. A final orbit correction removes the orbit-visible antisymmetric residue collectively, reaching the target in about three passes instead of eight (green star: five-seed ensemble, median 0.07×, all sub-target).

signed, bilinear quantity that scatters from seed to seed (Fig. 2), the robust statement is suppression of the whole ensemble below the target rather than any single number. As in any beam-based alignment, the reproducibility of each campaign — the stability of the magnetic centers under the gradient modulation used to find them — is a hardware input that sets how often the procedure must be repeated; quantifying it is outside the scope of this tracking study. The air-core (iron-free) quadrupoles foreseen for the experiment, being free of magnetic hysteresis, are favorable in this respect.

VI. OPERATING THE SCHEME: DRIFT AND THE TIME BUDGET

An alignment result is useful only if it survives on the machine. Quadrupole positions drift — thermally and with ground motion [9] — continuously re-loading random misalignment onto the corrected state. We simulated this by adding fresh random offsets of rms σ_d and re-measuring the false EDM.

The distinction that governs the operational picture is between the two misalignment components. Single-beam orbit feedback removes only the antisymmetric part of

TABLE II. The suppression chain at $\sigma = 10$ μm (single beam, tracking). Rows 2 and 4 are five-seed ensembles: row 2 the median of the counter-rotating–beam difference after orbit correction (spread 0.6–72×), and the last row the distribution of the full scheme (fixed rcond = 0.01, all seeds sub-target, median 0.07×); the raw and BBA rows follow the representative pattern (raw 356×). The equivalent EDM uses 1 nrad/s ↔ 10^{-29} $e$cm.

Stage	$\lvert f\rvert$ / target	EDM [$e$cm]
raw	356×	$\sim 4\times10^{-27}$
single-beam orbit corr. + CW/CCW	$\sim$ 47×	$\sim 5\times10^{-28}$
BBA (measured steering)	17–28×	$\sim 2\times10^{-28}$
BBA + final orbit corr.	0.03–0.22×	$\lesssim 10^{-29}$

the drift, so with it alone the false EDM stays below target for σ_d up to ~ 5 μm (0.05× at 2 μm, 0.3–0.6× at 5 μm), but the slowly accumulating *symmetric* part is invisible to it and, in the reference program, forces the beam-based alignment to be repeated. This is where counter-rotating correction earns its keep. As shown in Sec. V, the two-beam correction reaches the symmetric component — and, crucially for drift, it does so on the differential orbit without any absolute calibration: the drift is $\Delta q = q(t) - q(t_0)$, the corresponding orbit change $\Delta y = R\,\Delta q$ is free of the static BPM offset (which cancels between the two times), and its counter-rotating sum exposes the symmetric part of Δq. Because the offset cancels identically, the differential correction vector is the *same* as the offset-free one — so the tracked performance is exactly the offset-free ensemble of Table I, not a separate approximation — and each cycle re-references, so the cancellation does not accumulate error over a run. A continuous two-beam feedback on Δy thus holds the *symmetric* drift, not merely the antisymmetric one, at that offset-free level. The initial absolute alignment still requires one campaign — BBA, or the polarity four-fold — to set the symmetric component against the offset degeneracy of Sec. V; but its *maintenance*, the repeated re-alignment that the symmetric drift would otherwise demand, is replaced by the continuous feedback.

The time budget follows. The initial alignment is a one-time cost — a few-pass BBA campaign closed by an orbit correction is of order half a day (47 magnets × 2 planes × 2 beam positions × 2 gradient settings ≈ 400 orbit acquisitions per pass, ~ 10 s each) — after which the two-beam feedback runs during data taking at no extra cost and no further campaigns are scheduled for the symmetric drift. The dominant time scale of the experiment then remains the statistical one: polarimetry reaches the nrad/s level only over months to a year of running [10], and the continuous correction adds nothing to it. A word on the assumption that the BPM offset is static. The differential cancels the offset that is common to two successive readings, so only the offset *change* between them survives, and that change — a slow thermal or electronic drift — leaks into the symmetric channel like a small misalignment. In continuous (rolling) operation these per-cycle leaks accumulate as a random walk: the symmetric residue after N cycles is $\approx 0.96\,\dot{b}\,\tau_c\sqrt{N}$, where $\dot{b}$ is the offset drift rate and τ_c the cycle time (the coefficient 0.96 is the same offset-to-symmetric projection as above, and we confirm the $\sqrt{N}$ scaling in the tracker). Numerically, an offset drift of 1 μm/day corrected once per minute leaks only ~ 0.03 μm over a day and ~ 0.5 μm — the coherent-optics floor — over a full year of continuous running; a *static* offset of any size cancels exactly. The requirement is thus not that the offset be small, but that it drift slowly compared with the correction cadence, which realistic BPMs satisfy by orders of magnitude. Two hardware inputs not modeled here remain to quantify: the magnetic-center stability under the gradient modulation of the initial BBA (its repeatability), and a per-monitor BPM gain error, which enters the differential response like a β-beat (1% gain leaving ~ 0.6 μm of symmetric residue, so percent-level gain calibration suffices).

VII. CONCLUSIONS

The control program proposed for the symmetric-hybrid proton EDM ring pairs beam-based alignment with a four-fold measurement — the counter-rotating difference at both quadrupole polarities — that cancels the quadrupole-misalignment false EDM [1]. Using symplectic spin tracking we have asked how far orbit correction alone can carry that task, and the answer is set by the bilinear four-channel structure of the false EDM and by the sharply different orbit visibility of its two misalignment components.

(1) The false EDM is a bilinear functional of the two transverse misalignment patterns [Eq. (8)], verified in tracking at the 0.1% level, and it separates into an orbit-visible antisymmetric part and a symmetric part to which the orbit response is weaker by up to two orders of magnitude, mode by mode. It is therefore not characterized by an rms tolerance: at fixed σ it varies by more than an order of magnitude with the pattern, and a correction procedure changes not the size but the *component* in which the residual lies.

(2) The reference solution reaches the target but at an operational cost. Null-finding BBA reaches the symmetric component that orbit inversion cannot — it locates each magnetic center by a local zero crossing, not a global inverse — and, iterated far enough, drives the false EDM below target on its own (monotonically, to 0.09× by the eighth pass; there is no plateau), though slowly; a single final orbit correction removes the orbit-visible antisymmetric residue collectively and reaches the target in about three passes instead. BBA requires a faithful *horizontal* steering model, which an idealized lattice that neglects the horizontal focusing of the electric deflectors does not provide. The polarity four-fold reaches the floor in one step, but only if the machine presents the same symmetric alignment in two sequential polarity states; a 1 μm symmetric inter-configuration drift, invisible to the orbit, re-opens the false EDM to tens of times the target.

(3) Counter-rotating orbit correction reaches the symmetric component from the orbit, subject to one degeneracy. A single-beam measurement cannot: the symmetric misalignment is nearly invisible to it, and the counter-rotating *separation* the reference chain monitors is no better, being the *difference* of two nearly opposite orbits. Their *sum*, however, exposes the symmetric component fully (symmetric visibility 1.0 against 0.2 for the separation), and correcting both orbits jointly against a calibrated reference uses it: on the raw 10 μm machine the counter-rotating difference falls to a median 0.043× the target over fifteen seeds at a realistic 1% β-beat, all sub-target, unchanged by β-beating and a 0.2 mrad unmodeled quadrupole roll. The same sum, however, carries the static BPM offset, degenerate there with the symmetric misalignment (100 μm uncalibrated offsets return the remnant to $\sim 200\times$); so two-beam correction does not remove the need for an absolute reference, which one BBA campaign, or the four-fold, must still set.

(4) What it removes is the *periodic* realignment, and this is its operational role. Drift is differential: the static offset cancels in the orbit *change* $\Delta y = R\,\Delta q$, whose counter-rotating sum exposes the symmetric part of the drift with no offset contamination, so a continuous two-beam feedback holds the symmetric alignment — the component a single-beam feedback cannot see, and that in the reference program forces the BBA to be repeated — at the offset-free level indefinitely. The initial alignment is a one-time cost; its maintenance is then continuous and free, and the time budget is dominated by the statistical scale of the polarimetry [10]. The remaining items to quantify are the coherent optics floor (against which the method is robust to $\sim 1\%$ β-beat) and a per-monitor BPM gain error, which enters the differential response like a β-beat — a 1% gain adding ~ 0.6 μm of symmetric residue, comparable to the optics floor, so BPM gains calibrated to the standard percent level suffice.

All numbers refer to the symmetric-hybrid ring designed for the proton EDM experiment [1], in a linear model without sextupoles. A distinction between the qualitative and the quantitative should be kept in mind. The structural conclusions — the visibility split, the bilinear channel decomposition, and the exposure of the symmetric component in the counter-rotating sum rather than the difference — rest on the alternating-gradient focusing common to strong-focusing rings, and we expect them to carry over to other such lattices. The numerical values that quantify them — the symmetric visibilities (0.19 for the difference, 1.01 for the sum), the condition numbers (135 single-beam, 73 stacked), the direction-common residue (21%), and the suppression factors — are properties of this lattice, and would be recomputed for another; it is the structure, not these numbers, that we claim to be general. The principal open items are the magnetic-center stability of any BBA cross-check under gradient modulation (the hardware floor, not modeled here) and the verification on other lattice variants.

The tracking and analysis code, the response matrices, and the data behind the figures and tables are available from the authors on reasonable request.

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